Vibration Lab Report

ENG 4137: Vibration 4

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Introduction

This report presents an intensively comparative focused study of modal analysis using Experimental Modal Analysis (EMA) and Finite Element Analysis (FEA). It outlines the procedures for both methods, examines frequency response functions and mode shapes, and evaluates discrepancies through data interpretation, resonance behaviour, and analysis of limitations in each approach.

Methodology

To effectively compare the modal analysis results between EMA and FEA, it is crucial to understand the differences in methodology, including the modal parameter extraction process and the fundamental distinctions between the two approaches, overlaying the numerical and experimental results.

EMA

In EMA, modal parameter extraction is a procedure of identifying the key modal properties of a structure, such as natural frequencies, mode shapes, and damping ratios from the measured vibration data. The main procedures are as follows:

1. Data Acquisition (Excitation and Measurement)

A generator known as Dynamic Signal Analyser(Quattro) was connected to the pc and the accelerometer was positioned on top of the wing component at 10 different points. The plane was excited using a shaker attached to the response force transducer. Out of various excitation mechanisms such as random, burst rand, impulse, swept square, etc, a swept sine wave was used in the EMA lab because it ensures a highly accurate and efficient way to compute the frequency response function(FRF). The sine wave allows to sweep effectively from the wanted frequency of 10 to 500Hz. It also allows to obtain high signal to noise ratio(SNR) in which the response signal has minimal effect from noise since only one frequency is excited at a time. Lastly, signal leakage is avoided unlike random or impulse excitations, a swept sine wave provides well-controlled spectral content, reducing errors in Fourier Transform calculations.

2. Transduction System/Signal Processing

A transducer in this case ,an accelerometer, was attached on 10 positions of the wings to measure inputs and outputs. This step measures the input(force) and output(response) signals to compute FRF using ‘Coherence Function’ to ensure both signals are well-correlated, reducing noise effects. Lastly, a conditional amplifier was utilized to strengthen the signal.

3. Modal Parameter Estimation

Once the vibration data is imported using ME’Scope files, curve fitting and peak extraction were carried out to minimise the error by continuously refining the estimates by calculating the least squared error. Under Mode indicator, ‘Modal Peaks Function’ as well as ‘Modal Indicator Function’ were selected to compute modal analysis. Using ‘cursor Band & Count Peaks’ allowed to isolate and smoothen the modal peaks in the frequency spectrum. It used ‘Ortho Polynomial Method’ to estimate the frequency & damping ratios. Lasty, computed residues to finalize modal extraction.

4. Mode Shape Extraction

‘Sign Dwell Animation’ was selected to visualize the mode shapes.

FEA

In FEA using Abaqus, modal parameter extraction procedure follows a structured process similar to EMA, however, relies on numerical computations rather than the experimental data collection. The aim is the same to extract natural frequencies, and mode shapes in the set environment.

1. Model Setup

The plane was modelled by creating 4 components(Fuselage, Nose, Wing, Tail). Then the material properties were defined to the given values for Young's modulus, density, Poisson's ratio. The structure is discretized into finite elements with the global seed size of 0.02. Constraints were then applied accordingly.

2. Free Vibration Modal Analysis

Unlike EMA, FEA creates a function called 'Step' to computationally simulate the frequency response function(FRF). 'Linear Perturbation -Frequency' was selected for Step 1 to extract and analyse the natural frequencies and mode shapes for a free vibration.

3. Forced Vibration Analysis

In EMA, a swept sine wave is used to excite a structure across a range of frequencies sequentially, similarly, an equivalent procedure in FEA is Steady-State Dynamics analysis with a frequency sweep. 'Steady-state dynamics, Direct' was selected for Step 2 with applied over the given frequency range. The load in the Y direction is equivalent to exciting the structure using a single excitation point similar to an EMA test where the shaker applies the force. Then, the vertical response of the selected wingtip over the entire frequency is examined by creating a XY plot where x represents frequency, and y represents the displacement in log.

Results Comparison between EMA and FEA

Although EMA and FEA both aim to analyse the natural frequencies, and mode shapes of the structure, they are meant to produce slightly or potentially significantly different results due to several factors.

EMA is based on real-world physical measurements and data, capturing all boundary conditions, material properties and no assumptions are made. It naturally accounts for all forms of damping, including material, structural, and external damping. In addition, results depend on the accuracy of the excitation method and sensor placement, and captures real-world nonlinearities such as contact, and friction.

On the other hand, FEA utilizes an idealized mathematical model, assuming perfect boundary conditions, isotropic materials, no imperfections, and homogeneity. The material properties and boundary conditions are all assumed homogeneous and predefined. Results depend on the mesh density and element type as coarser meshes can miss finer vibrational details however increases the computational efficiency, hence finding the balance is critical to simulate. It does not suffer from measurement errors unless the parameters and conditions are not well-defined.

Harmonic Animation comparison

The FEA results in this section have been taken from the results obtained in the free vibration(Step 1) to identify the natural frequencies of the plane model, not influenced by any external force(Step 2). This provides insight into the plane model's inherent natural frequencies.

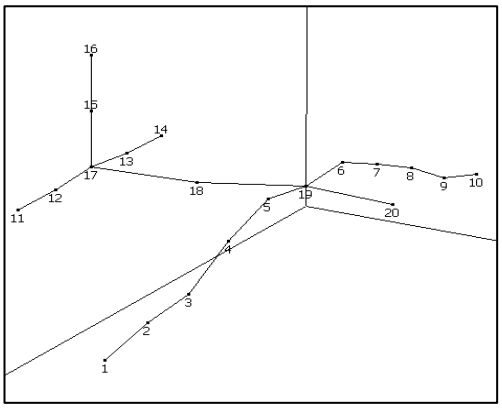


Figure 1: EMA Frequency: 29.28 Hz

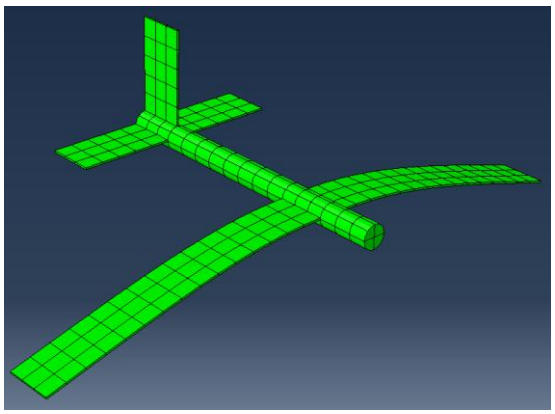


Figure 2: FEA Frequency: 21.048 Hz

The figures above illustrate the mode shapes at their respective frequencies. The mode shape captured in figure 2 correspond to the first peak in figure 9 of the frequency response function of the right wingtip. The FEA result represents the first mode shape from Step 1, showing strong agreement with the EMA results in terms of overall deformation patterns. The mode shapes align closely, particularly in their dominant features at this frequency. However, notable discrepancies are observed at nodes 3, 4, 8, and 9, where the vertical displacement in the EMA results exhibits abrupt drops, whereas the FEA simulation suggests a smooth, negative quadratic-like trend.

These differences may be attributed to several factors, including:

- The mode shapes being compared at different frequencies (29.28 Hz for EMA and 21.048 Hz for FEA), which can influence deformation characteristics.
- The influence of damping effects in EMA, which may not be fully accounted for in the FEA simulation.
- The presence of holes in the actual structure might reduce stiffness and alter mass distribution, potentially lowering natural frequencies and causing local deformation differences between EMA and FEA results.

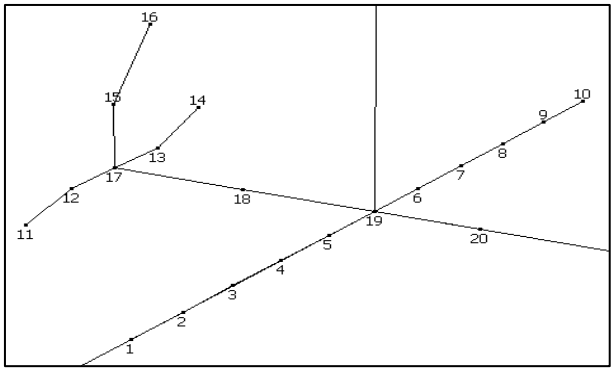


Figure 3: EMA Frequency: 169.69 Hz

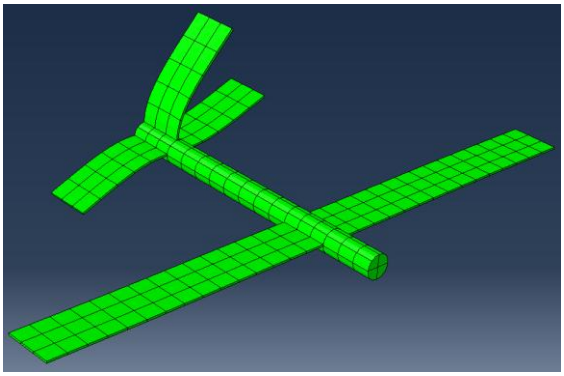


Figure 4: FEA Frequency: 170.56 Hz

The mode shape shown in Figure 4 at 170.56 Hz does not correspond to any resonance peak observed in Figure 9. This is expected, as Step 1 identifies the natural frequencies of the system, but this does not imply that all mode shapes will be excited or appear in the forced vibration results of Step 2. The frequency response function at the right wingtip (Figure 9) indicates no significant excitation at 170.56 Hz, likely because the response was measured at a single point, which may not be sensitive to that particular mode.

The figures above showcase the other mode shapes at similar frequencies, highlighting strong similarities between the two results. The wing remains flat, while the node 16 of the tail tilts towards the right and node 11 and 14 in EMA resemble the displacements in FEA, further reinforcing the correlation between the two methods.

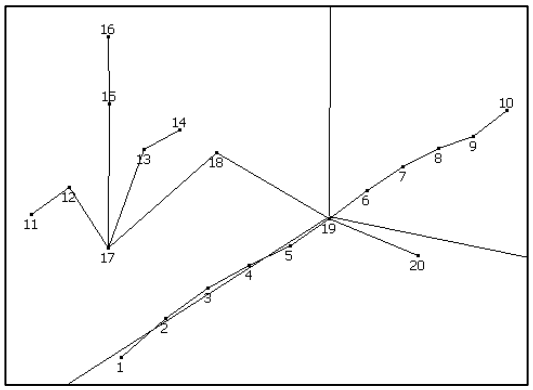


Figure 5: EMA Frequency: 214.95 Hz

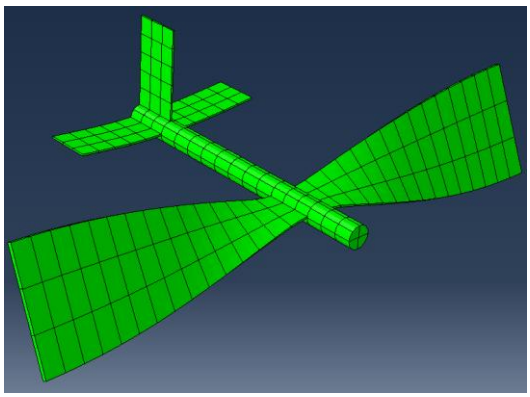


Figure 6: FEA Frequency: 214.77 Hz

The mode shape captured in figure 6 correspond to the 4th peak in figure 9. The figures above depict similar mode shapes. However, nodes 11, 12, 13, and 14 on the tail exhibit an abrupt incline in displacement in EMA, whereas FEA predicts only minor displacement in the same direction in this region.

From this frequency onward, we can reasonably assume that the increasing frequency contributes to these discrepancies. To further validate this hypothesis, we will analyse even higher frequencies.

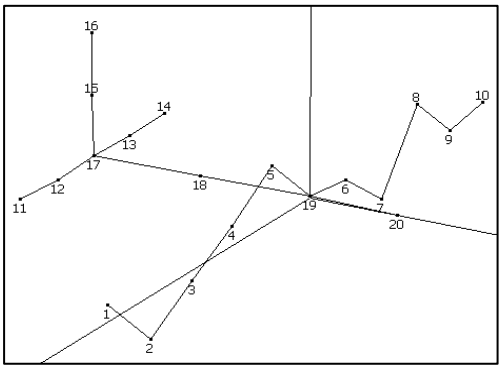


Figure 7: EMA Frequency: 325.71 Hz

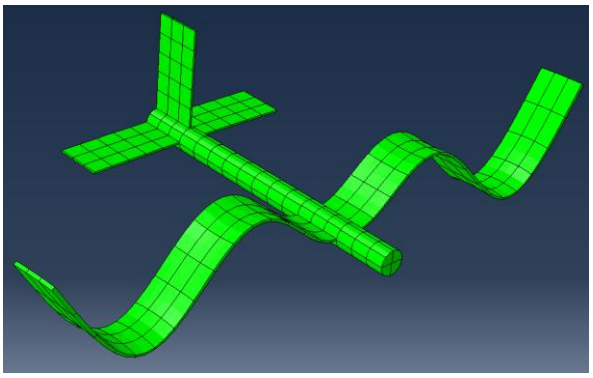


Figure 8: FEA Frequency: 328.39 Hz

The mode shape captured in figure 8 correspond to the 6th peak in figure 9. The figures above illustrate significant differences in the wing's mode shape. While the FEA model exhibits a symmetrical wing deformation, the EMA results display irregular and unpredictable wing displacement, lacking a clear pattern. However, this does not imply inaccuracy, as the EMA results could still represent the true structural behaviour. This could be due to analysing the mode shapes at higher frequencies. At higher frequencies, mode shapes become more complex and sensitive to small variations in geometry, mass distribution, and boundary conditions. Minor differences in the FEA model(omission of holes, undamped) can lead to larger discrepancies in mode shapes.

At lower frequencies, global deformation dominates as the structure behaves more as a whole, whereas at higher frequencies, local deformations become more significant, making small mismatches more noticeable. In addition, the inhomogeneity in terms of material properties in EMA could also make significant differences when compared to FEA results.

Modal Analysis Comparison

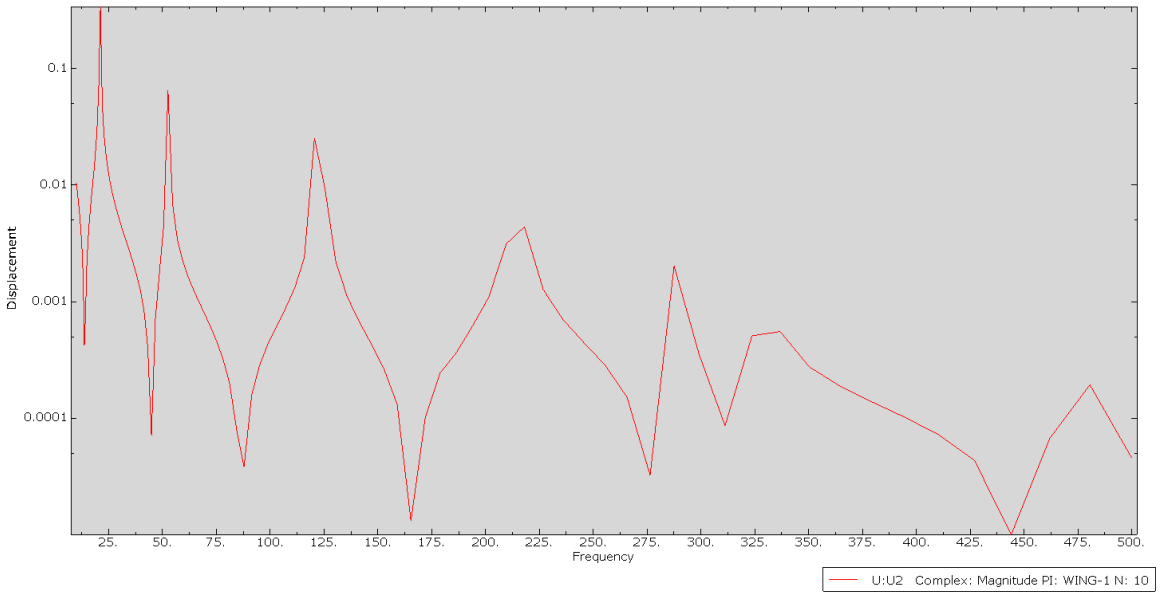


Figure 9: FEA Frequency Response Function of the right wingtip

The figure 9 depicts a decreasing resonance amplitude response with frequency due to increasing system stiffness. Since the displacement response is inversely proportional to stiffness at higher frequencies, the system becomes stiffer, leading to a reduction in displacement amplitude despite the absence of damping. Moreover, higher-frequency modes are often more localized, meaning the energy distribution shifts compared to lower modes. With no damping, higher-frequency modes still follow linear behaviour, where energy is spread across multiple nodes. A more detailed comparison with the Experimental Modal Analysis (EMA) results will be presented later in the report.

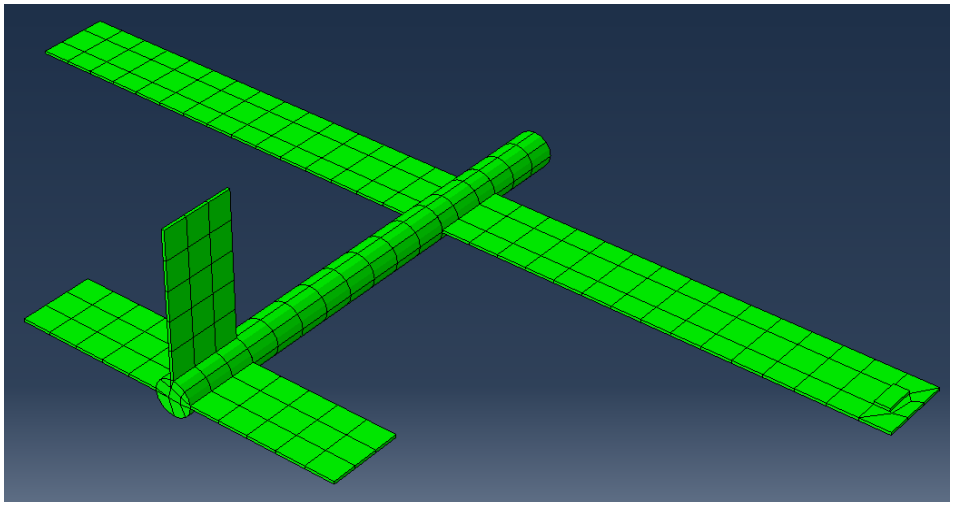


Figure 10: Addition of a mass on the right wing edge corresponding to FRFs in Figure 11

The figure 10 illustrates the addition of a small mass on the right wing edge, representing a sensor or accelerometer used in EMA. The mass dimensions are 0.01 x 0.02 x 0.002 m^3 and appropriate partitioning was applied to ensure accurate meshing and analysis of the wing model.

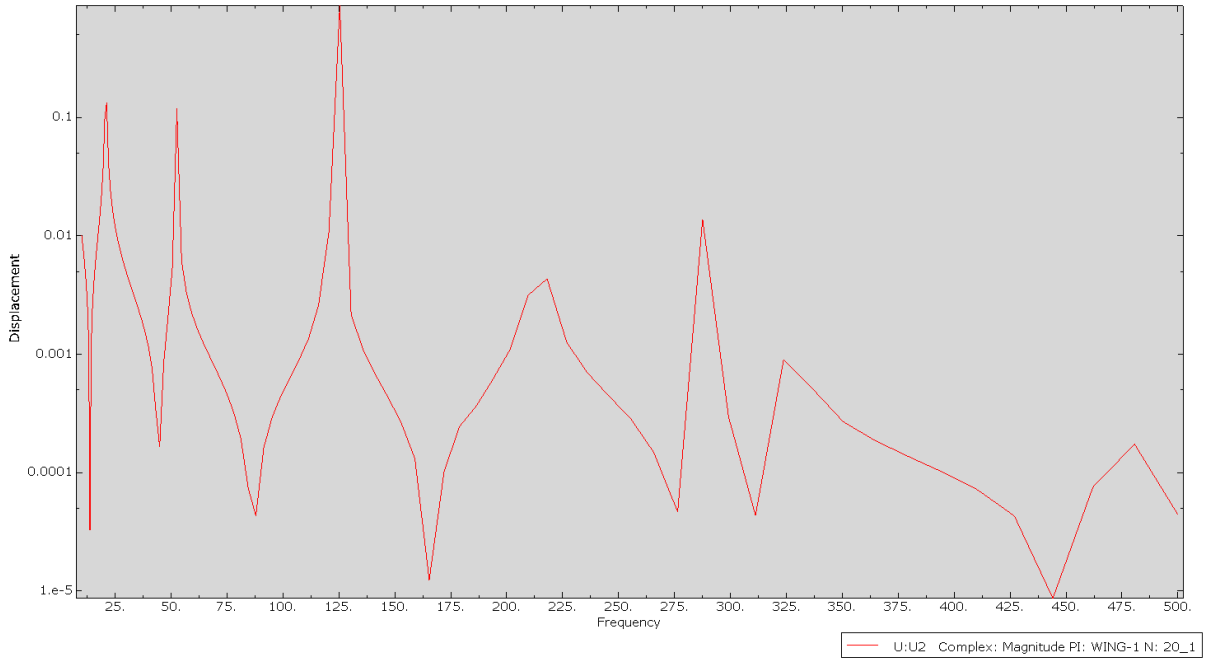


Figure 11: FEA Frequency Response Function of the right wingtip with additional mass

As shown in the figure above, although the accelerometer represents a small mass, it can still influence the dynamic behaviour of the system. When added to a flexible structure like the wing, it locally alters the mass distribution. Notably, the third and fifth resonance peaks have significantly increased in amplitude compared to the FRF shown in Figure 9, while the total number of resonance peaks remains constant at seven. The remaining peaks exhibit similar magnitudes and trends. This suggests that the third and fifth modes are particularly sensitive to mass at the accelerometer location, resulting in amplified responses at those frequencies. This highlights the sensitivity of the FRF to both the measurement and excitation locations.

Additionally, the accelerometer introduces not only mass but also localized stiffness, especially when rigidly attached to the wing edge as a homogeneous extension of the structure -potentially leading to minor stiffness discontinuities.

It is important to note that this analysis was conducted solely to understand the influence of a small added mass on the dynamic response. For the remainder of the report, analyses will proceed without the added mass to preserve symmetry and simplify interpretation, especially as the FRF measurement point varies across cases which will be covered more thoroughly. Nonetheless, this section demonstrates the notable effect a small mass can have on system behaviour.

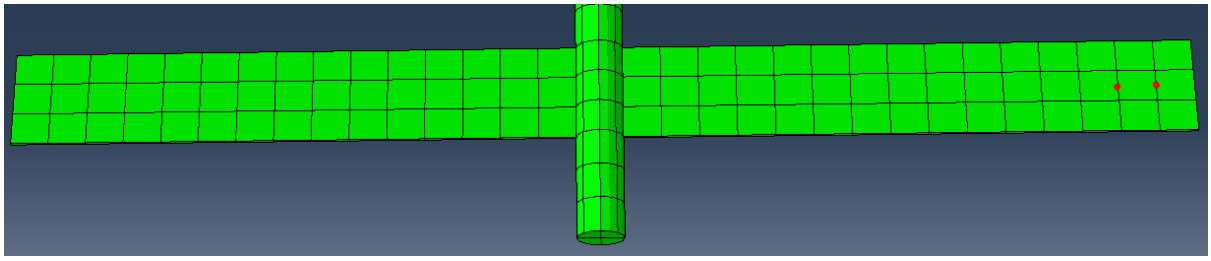


Figure 12: Locations of the 2 selected points corresponding to FRFs in Figure 13

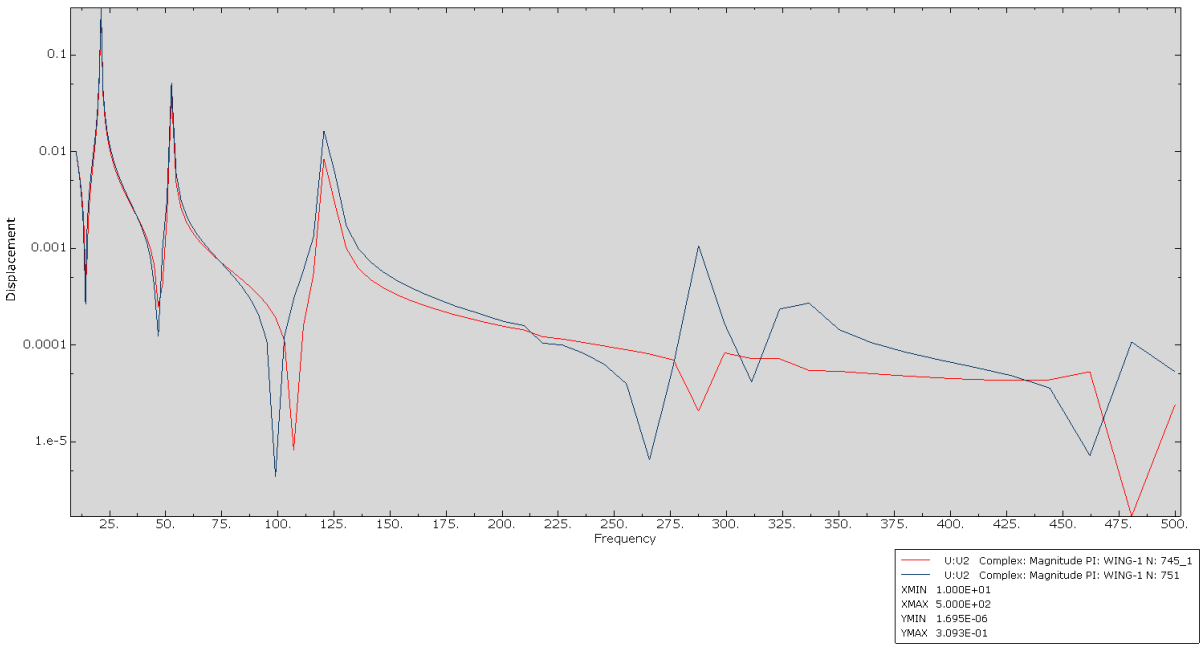


Figure 13: FEA Frequency Response Function at 2 different points

Figure 13 presents two different Frequency Response Functions (FRFs) obtained from different points on the structure, as indicated in Figure 12. Despite the proximity of these points, the displacement response exhibits significant variation as frequency increases. Up to 200 Hz, both graphs follow a similar trend; however, beyond this threshold, their responses diverge completely, demonstrating different resonance peaks.

As previously discussed, higher-frequency mode shapes are increasingly sensitive to small variations in geometry, mass distribution, and boundary conditions. The fact that two closely spaced points yield substantially different results highlights this sensitivity. Consequently, at higher frequencies, EMA and FEA would exhibit significant discrepancies, as each method operates under different assumptions and constraints within the modal analysis framework.

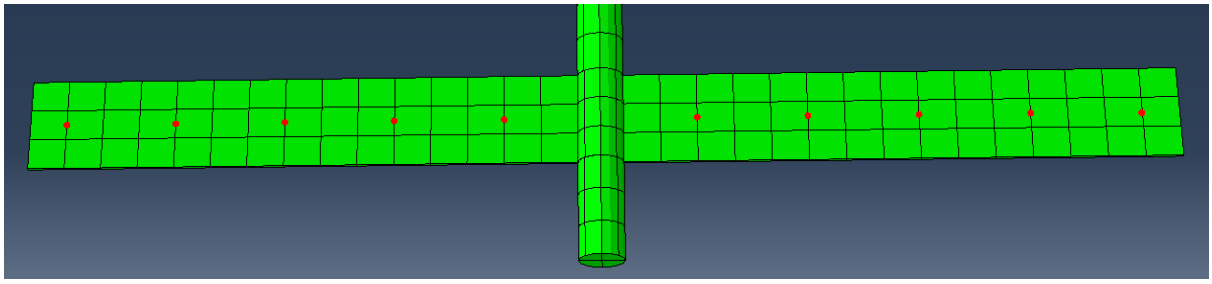


Figure 14: Locations of the 10 selected points corresponding to FRFs in Figure 15

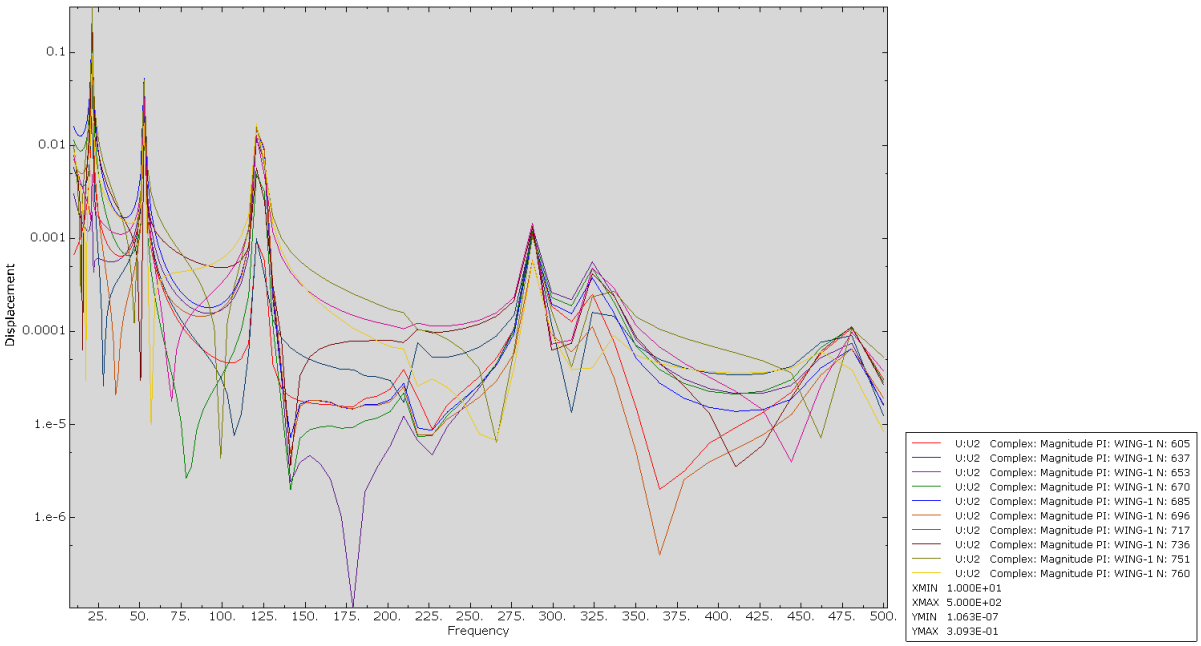


Figure 15: FEA Frequency Response Function at 10 different points

Figure 15 presents 10 different Frequency Response Functions (FRFs) obtained from 10 different points on the structure, as indicated in Figure 14. These points correspond to the same node locations as those in the actual plane model, ensuring a valid comparison with EMA.

Figure 15 exhibits six distinct resonance peaks, consistently observed across all 10 nodes, indicating the most dominant natural frequencies of the structure. In contrast, Figure 9 displays seven resonance peaks, highlighting a discrepancy in the identified modal responses. This difference can be attributed to the selection of measurement points, as Figure 9 captures the response at the right wingtip, whereas Figure 15 considers 10 points within the wing. The response measured at different locations is influenced by local stiffness variations. The wing tip, being a free end, may exhibit additional resonance peak compared to interior points of the wing, where the motion is more constrained. To ensure a more accurate comparison between EMA and FEA, Figure 15 is the more relevant reference, as the accelerometer in the EMA setup was positioned within the wing.

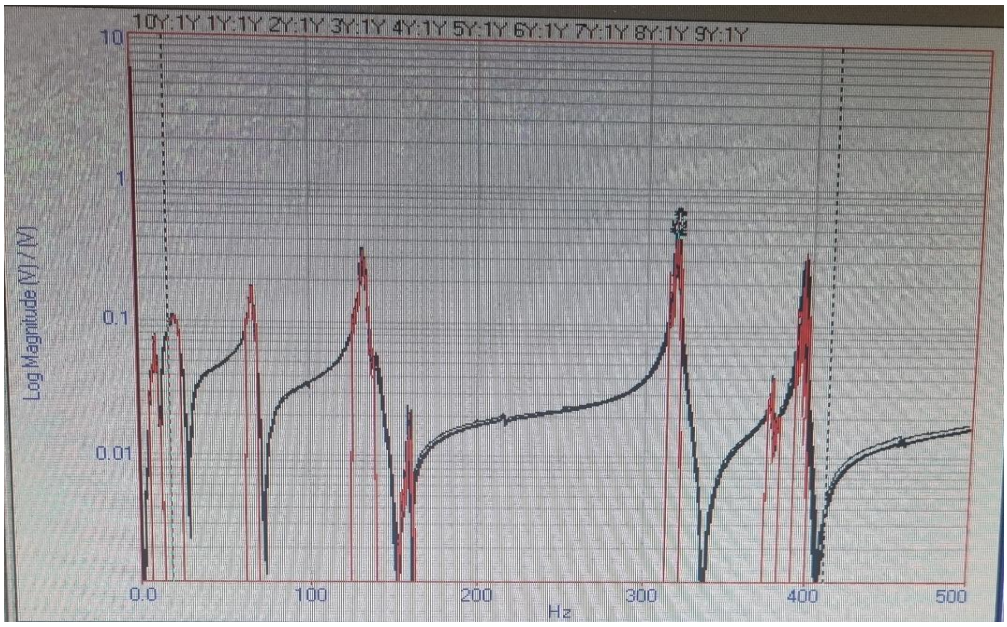


Figure 16: EMA Frequency Response Function at 10 different points

Figure 16 presents the EMA FRF at 10 different points. It is immediately recognizable that the FRFs in EMA and FEA portray quite different trends. The natural frequencies observed in EMA are approximately 18 Hz, 23 Hz, 75 Hz, 130 Hz, 160 Hz, 320 Hz, 375 Hz and 390 Hz, a total of 8 eigenfrequencies. Whereas in FEA, they are 23 Hz, 52 Hz, 125 Hz, 283 Hz, 325 Hz, and 480 Hz, a total of 6 eigenfrequencies shown in figure 15.

Moreover, EMA results display a higher-frequency modes showing larger peak displacements. This could be due to several reasons as follows:

In an excited system, the transmissibility of applied force increases with frequency, leading to greater response amplitudes near resonance. Unlike in FEA, real-world excitation in EMA can amplify certain higher-frequency responses, especially if the excitation source excites higher modes effectively. Higher modes tend to have localized deformations, meaning displacement concentrates in specific regions of the structure. As discussed earlier, EMA possesses realistic damping behaving non-linearly. At lower frequencies, damping effects might be more pronounced, reducing peak amplitudes and damping effects may be weaker at higher frequencies.

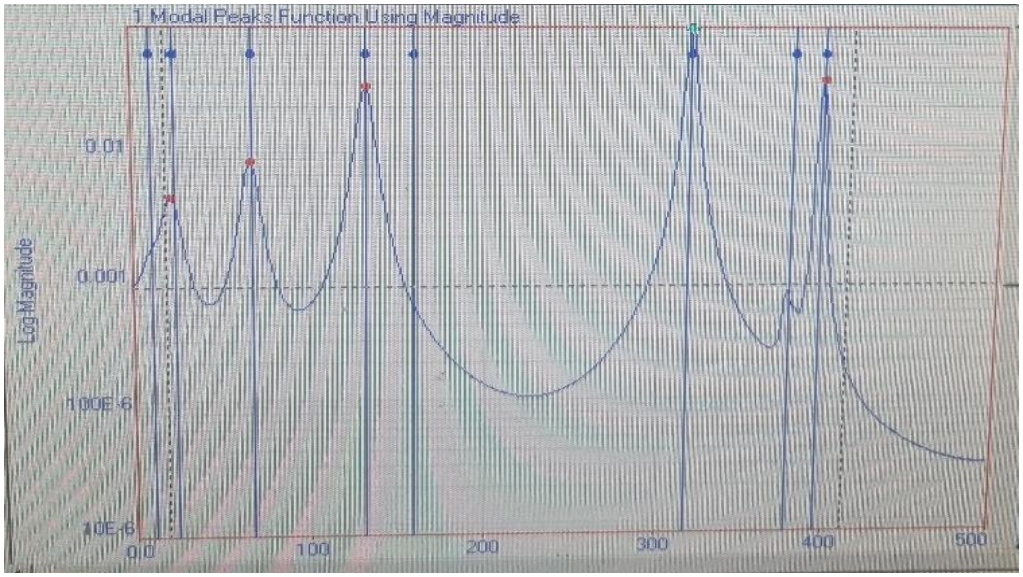


Figure 17: EMA Modal Peaks Function at 10 different points

The Modal Peaks Function(MPF) in EMA serves several key purposes in identifying and analysing the dynamic behaviour of a structure. Resonance peaks in FRF correspond to the natural frequencies, and the MPF helps pinpoint these frequencies as shown in figure 17, displaying those in red or blue dots. The red and blue dots indicate difference confidence levels or significance of detected modal peaks. Red dots represent confirmed modal peaks that have been identified as dominant natural frequencies or strong resonance peaks which usually have a high signal-to-noise ratio. Whereas blue dots represent potential/secondary mode peaks that are less prominent or uncertain due to lower amplitude, noise interference, or measurement inconsistencies.

Figure 16 illustrates that eight eigenfrequencies were identified in the EMA results; however, through Modal Peaks Function (MPF) analysis, only four significant resonance peaks were detected at 23 Hz, 75 Hz, 130 Hz, and 390 Hz, each corresponding to distinct mode shapes.

In comparison, the FEA results in Figure 15 identified six eigenfrequencies at 23 Hz, 52 Hz, 125 Hz, 283 Hz, 325 Hz, and 480 Hz. The closest matching frequencies between EMA and FEA are observed at 23 Hz and within the range of 125–130 Hz, suggesting that the mode shapes at these frequencies are strongly correlated and represent true structural resonance modes where other nodes might potentially suffer from local stiffness effects. Additionally, it is highly possible that similar errors could be incurred if the maximum frequency has been extended from 500 to higher where there will be remarkably apparent mismatches present between the FEA and the EMA.

However, this does not imply that the other natural frequencies identified in EMA and FEA are irrelevant, but rather that they may be influenced by factors such as measurement sensitivity, damping effects, or modelling simplifications. Further investigation may be required to validate these additional modes which would be covered in limitations.

Limitations

There are several limitations to consider when comparing EMA and FEA in modal analysis. While some of these have already been identified and discussed earlier in the report, others will be addressed in detail in this section to provide a more comprehensive understanding of the factors influencing the results.

1. *Limited Sensor Coverage in EMA*

Only 10 accelerometer points were used, all placed on the wing. This provides an incomplete representation of full structural dynamics, especially for modes involving the fuselage, nose, or tail, which may have been missed or inaccurately captured. Due to limited lab time, the completeness and resolution of the experimental data set was affected.

2. Shaker Placement and Excitation Method in EMA

The shaker was not placed near the edge, but rather at the wingtip, which may have over-excited certain modes. This influences which mode shapes are effectively captured, especially those involving large displacements at or near the tip.

3. Mismatch Between EMA and FEA Sensor Locations

The points selected in FEA for FRF extraction may not precisely match the actual accelerometer positions in EMA, leading to discrepancies in the comparison due to differences in modal sensitivity at different locations.

4. Mesh Density and Element Type in FEA

The finite element mesh resolution affects the accuracy of mode shape predictions. A coarser mesh may not capture higher-frequency modes accurately, while a finer mesh increases computational time, and was possibly not fully optimized. The global seed size could be decreased instead of a fixed value of 0.02 to ensure a finer mesh but at the cost of reduced computational efficiency.

5. Material and Geometry Simplifications in FEA

The FEA model does not account for holes, joints, or imperfections present in the real structure. These omissions lead to differences in stiffness and mass distribution, affecting natural frequency and mode shape predictions. Frankly, the FEA plane model could be extruded in the area of the holes to more accurately capture the comparison and ensure consistency of the controlled variable.

6. Lack of Damping in FEA

The FEA model does not include realistic damping, while EMA captures material, structural, and environmental damping. This affects both the amplitude and shape of resonance peaks, especially at higher frequencies.

7. Boundary Conditions

The real plane model’s mounting conditions (glue tags or supports) may introduce unwanted weight, stiffness or compliance not captured in the FEA model, leading to frequency and shape mismatches.

8. Use of Swept Sine Excitation Only

While swept sine waves provide accurate and low in noise results, other excitation types (random or burst random) might excite a broader range of modes. The selected method may have missed certain higher modes or complex coupling.

9. Lack of Coherence analysis in FEA

Coherence analysis between FRFs and a correlation study between mode shapes could help validate and enhance the agreement between EMA and FEA results.

10. Mass Loading Effective of the Accelerometer in EMA

Although the accelerometer has a small mass, its placement on the flexible wing structure introduces local mass loading, which can alter the dynamic behaviour. While this was briefly analysed earlier, its contribution adds uncertainty to the direct comparison between EMA and FEA, as the FEA model assumes a massless measurement system.

Conclusion

This report compared Experimental Modal Analysis and Finite Element Analysis in determining the natural frequencies, mode shapes, and dynamic behaviour of a structure. While EMA provides real-world accuracy, capturing actual damping effects and boundary conditions, FEA offers a controlled and idealized environment, allowing for predictive analysis before physical testing.

In the study, a single vibrating mechanism—a shaker—was used to excite a single vibrating system, represented by a plane model. This setup ensured that no coupled vibrations occurred, minimizing the complexity of the vibration patterns. Consequently, the absence of interaction effects between multiple vibrating systems contributed to more consistent and reliable data acquisition for the comparison.

Both methods are essential—EMA validates numerical models, while FEA enables efficient structural optimization. Despite discrepancies, their combined use enhances engineering accuracy and reliability. Future improvements in measurement techniques and modelling refinements can further bridge the gap, ensuring more precise structural analysis and design validation.

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